

KOSHIKA APEX CELLULAR THERAPY & BMT CENTRE

Clinical Diagnostic Division & Transplant Core
StemBridge Technology Park, Electronic City, Bangalore - 560100

NABL ISO 15189 & FACT-JACIE AFFILIATED

Report ID: MED-KOSHIKA-2026-8849
Barcode: KOSH*0914*88049

BUSULFAN FIRST-DOSE PHARMACOKINETICS & AREA UNDER CURVE (AUC) MONITORING

Patient Name:	Clinical Patient #49 (Verified Cohort)	Date of Exam:	September 14, 2026
MRN / Patient ID:	KOSH-BMT-2026-1049	Age / Gender:	24 Yrs / Male
Blood Group & Rh:	B+	Referring Doctor:	Dr. Ramesh Sarin, MD, DM (Hemato-Oncology)
Clinical Diagnosis:	Hematologic Malignancy Protocol Assessment #49	Specimen / Source:	Serial Plasma Samples

Test / Assay Target	Observed Value	Reference Range	Method / Assay	Status / Interpretation
Calculated AUC0-inf (Dose 1)	1,140 mcM*min	900 - 1,350 Target	LC-MS/MS Kinetics	Optimal Exposure
Peak Concentration (Cmax)	1,220 ng/mL	Expected Range	Mass Spectrometry	Within Therapeutic Safe Limit
Clearance Rate (CL)	2.6 mL/min/kg	Normal Pediatric	Non-Compartmental	Normal Hepatic Clearance
Dose Modification Recommendation	Keep Current Dose (100%)	80 - 120 % Target	Clinical Pharmacokinetics	Zero Dose Change Required

CLINICAL INTERPRETATION & RECOMMENDATIONS:

- Busulfan exposure (AUC 1,140 mcM*min) falls precisely within optimal target therapeutic window.
- Recommendation: Continue planned intravenous Busulfan dose every 6 hours without alteration.
- All assay analytical internal and external controls passed standard quality criteria.
- Report electronically signed and validated under NABL laboratory oversight protocols.

Scientific Analyst Team
Medical Technologist / Pathologist

Dr. Transplant Pathologist, MD
Consultant Hematopathologist

Dr. Director of Clinical Services, MD, DM
Director of Cellular Therapy & BMT